SUPPLEMENT TO: REAL MULTIPLICATION AND THE GENERALIZED FERMAT EQUATION $x^5 + y^3 = z^p$

MANVIR JASWAL

This Supplement to the paper *Real multiplication and the generalized Fermat equation* $x^5 + y^3 = z^p$ contains, for $p \equiv 19, 29, 31, 41 \pmod{60}$, the reduction of the equation to rational points on modular curves and the computations on which it rests; its table of computations also lists item C22, used in §2.5 of the paper. Numbers without the prefix S refer to the paper. The results that use [10] name the statements of [10] they assume by the labels (LL), (Ex), (Ex$_0$), (EL), (T2.1), (P7.4) and (LI) of §2.5 of the paper. We use the notation of §6.1 of the paper.

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S1. The primes $p \equiv 19, 29, 31, 41 \pmod{60}$

For these primes, Theorem 6.21 of the paper leaves three classes of solutions. We prove two consequences of the absence of congruences with J_s (§S1.1), reduce the equation to rational points on three modular curves (§S1.2), and describe the reduction at p in the classes in which $\mathfrak{p}$ is inert in the field of complex multiplication (§S1.3).

Remark S1.1. Each class contains a trivial solution: $Z = 0$ ($t = 1$) for (E), $X = 0$ (J_0, with complex multiplication by $\mathbb{Q}(\zeta_5)$) for (Z), $Y = 0$ (J_∞, by $K(\sqrt{-3})$) for (T). We expect that at 2, 3, $\sqrt{5}$ and ∞ the solutions of a class have the local representations of its trivial solution; and an auxiliary prime $\ell \nmid 30$ may divide Z, X or Y respectively. So we do not expect local comparisons, or auxiliary primes in the style of Mazur, to exclude any class. Excluding (E) calls for an isogeny theorem in the style of [7], and (Z), (T) for an argument in the style of [1]; Theorem S1.12 isolates what is needed.

S1.1. **Two consequences of the absence of congruences with J_s.** For $p \equiv 19, 29, 31, 41 \pmod{60}$, given (P7.4)(a) at $\ell = 3$, $\bar{\rho}_{t,\mathfrak{p}}$ is congruent to no J_s for any primitive solution (Corollary 5.2).

Remark S1.2 ([10, Theorem C] for every Y, for $p \equiv 19, 29, 31, 41 \pmod{60}$). Let (LI′) be the large-image hypothesis stated in Remark 6.16 of the paper. Assume (LL), (Ex), (EL), (T2.1) and (P7.4)(a) at $\ell = 3$; (LI) for $L = K' = K$, with constant C; and (LI′). Then for every prime $p \notin \mathcal{P}$ with $p > \max(C, C')$ and $p \equiv 19, 29, 31, 41 \pmod{60}$, the equation $X^5 + Y^3 = Z^p$ has no primitive solution; equivalently, $x^5 + y^p + z^3 = 0$ has no nontrivial primitive solution.

Proof of the statement in Remark S1.2. Let (X, Y, Z) be a primitive solution and $\mathfrak{p} \mid p$. By Mihăilescu's theorem [8], $Z \neq \pm 1$; let q be a prime dividing Z. Then J_t has potentially multiplicative reduction at the primes of K above q: by (T2.1) if $q \nmid 15$, by (P7.4)(a) at $\ell = 3$ if $q = 3$ (as $v_3(t-1) = p\, v_3(Z) > 3$), and by Lemma 2.19(b) if $q = 5$. Let $D = \mathrm{End}_{\overline{\mathbb{Q}}}(J_t) \otimes \mathbb{Q}$. Over a finite extension of the completion of K at such a prime, over which J_t is semistable and its endomorphisms over $\overline{\mathbb{Q}}$ are defined, D acts on the character group of the torus in the special fiber of the Néron model, which is nonzero of rank at most 2 [4, Exp. IX]. By Albert's classification [9, §21], D is $\mathbb{Q}$, a quadratic field, a quartic field, a quaternion division algebra over $\mathbb{Q}$, or one of $\mathbb{Q} \times \mathbb{Q}$, $\mathbb{Q} \times F$, $F \times F'$, $M_2(\mathbb{Q})$, $M_2(F)$, for imaginary quadratic fields F, F'. A quartic field, a quaternion division algebra and $M_2(F)$ have no nonzero module of $\mathbb{Q}$-dimension at most 2; and D contains K (Proposition 2.2), which embeds in none of $\mathbb{Q}$, F, $\mathbb{Q} \times \mathbb{Q}$, $\mathbb{Q} \times F$, $F \times F'$, and is the only quadratic field containing K. So D is K or $M_2(\mathbb{Q})$. If $D = K$, then $\mathrm{End}_K(J_t) \otimes \mathbb{Q} = K$ and (LI) applies to J_t over K; if $D \cong M_2(\mathbb{Q})$, (LI′) applies. As p splits in K, in both cases the image of $\bar{\rho}$ contains $\mathrm{SL}_2(\mathbb{F}_p)$. So $\bar{\rho}$ is irreducible, and it is not isomorphic to the reduction of a newform with complex multiplication by a field M, which is reducible on the subgroup G_M of index 2, because $\mathrm{SL}_2(\mathbb{F}_p)$ has no subgroup of index 2. By (LL), (Ex) and (EL) (Remark 2.14), $\bar{\rho}$ is congruent to some J_s, which Corollary 5.2 excludes. □

Theorem S1.3. *Assume* (LL) *and* (EL) *(in particular at level* $3^3(\sqrt{5})^3$*, with the identification of its two surviving forms with* $J_{-1/8}$ *and* $J_{9/8}$*), together with* (T2.1) *and* (P7.4)(a) *at* $\ell = 3$*. Let* $p \equiv 19, 29, 31, 41 \pmod{60}$ *with* $p \geq 139$*. Then no primitive solution* (X, Y, Z) *has* $\varepsilon_3 = \varepsilon_5 = 3$*, that is, has a Frey Jacobian* J_t *whose* λ*-adic representations over* K *have conductor exponent* 3 *at* (3) *and at* $(\sqrt{5})$ *(equivalently,* $J_t/\mathbb{Q}$ *has conductor exponent* 6 *at* 3 *and* 5 *at* 5*).*

Proof. (i) If $5 \mid Z$, then as in Lemma 6.3(b) $\rho|_{G_F}$ has exponent at most 2, and if $3 \mid Z$, then $v_3(t-1) = p\, v_3(Z) > 3$, so J_t has potentially multiplicative reduction at 3 by (P7.4)(a), and likewise exponent at most 2 there; so $3 \nmid Z$ and $5 \nmid Z$, and J_t has potentially good reduction at (3) and $(\sqrt{5})$ (Lemmas 6.2 and 6.3). (ii) As $\varepsilon_3 = 3$ is odd, $\bar{\rho}|_{I_3}$, hence $\bar{\rho}$, is absolutely irreducible (Lemma 6.1). (iii) By (i), inertia at (3), $(\sqrt{5})$ acts through finite groups of order prime to p (prime divisors q have $q - 1 \leq 4$), so the residual exponents are 3, and, with Theorem 2.8, [10, Theorem 7.8] gives $\bar{\rho} \cong \bar{\rho}_{g,\mathfrak{P}}$ with g of level $3^3(\sqrt{5})^3$. (iv) By the proof of [10, Theorem A], every newform of that level except the forms of $J_{-1/8}$ and $J_{9/8}$ is discarded for $p \notin \mathcal{P}$; so $\bar{\rho}$ is congruent to some J_s, which Corollary 5.2 excludes. □

Remark S1.4. Under (Ex_0), Theorem 6.21 already implies the conclusion of Theorem S1.3: in (E), $30 \mid Z$; in (Z), $3 \mid X$, so $\varepsilon_3 = 2$; in (T), $5 \mid Y$, so $\varepsilon_5 = 2$. The proof above uses the conductor of J_t directly and does not use (Ex_0).

S1.2. **A conditional theorem.** Here $p \equiv \pm 1 \pmod 5$, so $k_{\mathfrak{p}} = \mathbb{F}_p$. Over $\mathcal{U} = \mathbb{P}^1_K \smallsetminus \{0, 1, \infty\}$ the curves C_t form a smooth family (Section 2.2) with Jacobian $\mathcal{J}$; $\mathcal{J}[\mathfrak{p}]$ is finite étale of rank 2 over $\mathbb{F}_p$, and $\bar{\rho}_{t_0}$ is the action of G_L on its fiber at $t_0 \in \mathcal{U}(L)$, for a field $L \supseteq K$. In $G = \mathrm{GL}_2(\mathbb{F}_p)$ let B be a Borel subgroup, and $N_{\mathrm{sp}} \supset C_{\mathrm{sp}}$, $N_{\mathrm{ns}} \supset C_{\mathrm{ns}}$ split and non-split Cartan subgroups and their normalizers. For $H \subseteq G$ containing the scalars with $\det H = \mathbb{F}_p^\times$, let $\mathcal{Y}_H(\mathfrak{p})$ be the quotient by H of the $\mathcal{U}$-scheme of frames $\mathbb{F}_p^2 \xrightarrow{\sim} \mathcal{J}[\mathfrak{p}]$, and $\mathcal{X}_H(\mathfrak{p})$ its smooth projective model, with $t\colon \mathcal{X}_H(\mathfrak{p}) \to \mathbb{P}^1_K$. Write $\mathcal{X}_B = \mathcal{X}_B(\mathfrak{p})$ (the scheme of lines in $\mathcal{J}[\mathfrak{p}]$), $\mathcal{X}^+_{\mathrm{sp}} = \mathcal{X}_{N_{\mathrm{sp}}}(\mathfrak{p})$ and $\mathcal{X}^+_{\mathrm{ns}} = \mathcal{X}_{N_{\mathrm{ns}}}(\mathfrak{p})$.

Lemma S1.5. *Let λ be a prime of K, T_λ the λ-adic Tate module of the geometric generic fiber of $\mathcal{J}$, a free $\mathcal{O}_\lambda$-module of rank 2, and $\rho\colon \Pi \to \mathrm{GL}(T_\lambda)$ the action of $\Pi = \pi_1(\mathcal{U}_{\overline{K}})$, which commutes with $\iota(\mathcal{O}_K)$. By* [3, Exp. XIII, Corollaire 2.12] *there are $\gamma_0, \gamma_1, \gamma_\infty \in \Pi$ that generate inertia groups at points above $0, 1, \infty$, satisfy $\gamma_0\gamma_1\gamma_\infty = 1$, and generate Π topologically; fix such elements. Let $\zeta = \gamma_0(t^{1/5})/t^{1/5}$, a primitive fifth root of unity. Then $\det\rho = 1$, and:*

- *(a) $\rho(\gamma_0)$ has order 5 and characteristic polynomial $x^2 - \vartheta x + 1$, where $\vartheta \in \mathcal{O}_K$ is the root of $x^2 + x - 1$ such that $\iota(\vartheta)$ multiplies du/v by $\zeta^2 + \zeta^{-2}$; so $\vartheta = \zeta^2 + \zeta^{-2}$ if $\delta = 1$ and $\vartheta = \zeta + \zeta^{-1}$ if $\delta = -1$ (Proposition 2.2(b));*
- *(b) in some $\mathcal{O}_\lambda$-basis of T_λ, $\rho(\gamma_1) = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$;*
- *(c) $\rho(\gamma_\infty)$ has order 3 and characteristic polynomial $x^2 + x + 1$.*

Proof. For $i = 0, 1, \infty$, the restriction of ρ to an inertia group above i is the action of $G_\mathcal{K}$, $\mathcal{K} = \overline{K}((z))$ with $z = t, t-1, 1/t$, on the Tate module of the Jacobian J_t of the generic fiber over $\mathcal{K}$.

(a) As $\overline{K}((t))(t^{1/5})/\overline{K}((t))$ is totally ramified of degree 5, ζ is a primitive fifth root of unity. Let $\tau \in \overline{\mathcal{K}}$ with $\tau^5 = -t$; for $g \in G_\mathcal{K}$, $\zeta_g := g(\tau)/\tau = g(t^{1/5})/t^{1/5}$. By Lemma 2.1, $h_X(U,V) = (-X^2/U, -X^5V/U^3)$ is an isomorphism $C_{X,1} \to C_{-X^5}$, and for $\mu \in \mu_5$, $a_\mu(U,V) = (\mu U, \mu^3 V)$ maps $C_{X,1}$ to $C_{\mu^3 X,1}$, with $h_{\mu^3 X} \circ a_\mu = h_X$. The curve $V^2 = U^6 + 10\tau U^3 + 12U + 5\tau^2$ is smooth over $\overline{K}[[\tau]]$ (its discriminant $2^{12}3^6 5^5(\tau^5+1)^2$ is a unit), with special fiber $C_{0,1}$; as the torsion of its Jacobian is constant over $\overline{K}[[\tau]]$, h_τ and reduction identify $T_\lambda(J_t)$ with $T_\lambda(J_0)$, compatibly with ι, which on J_0 is the specialization of the real multiplication of this family (proof of Proposition 2.2). For $P \in C_{\tau,1}(\overline{\mathcal{K}})$, $g(h_\tau(P)) = h_{\zeta_g\tau}(g(P)) = h_\tau(a_\mu(g(P)))$ with $\mu = \zeta_g^{-2}$; so g, which acts trivially on the residue field, acts on $T_\lambda(J_0)$ through the automorphism $(U,V) \mapsto (\zeta_g^{-2}U, \zeta_g^{-1}V)$ of $C_{0,1}$ [4, Exp. IX, 4.2]. For $g = \gamma_0$ this is $\beta = \alpha^{-2}$, where $\alpha(U,V) = (\zeta U, \zeta^3 V)$. By the proof of Lemma 2.5(a), $\mathbb{Q}[\alpha] \cong \mathbb{Q}(\zeta_5)$ is a field with real subfield $\iota(K)$, and $V_\lambda(J_0)$ is free of rank one over $\mathbb{Q}[\alpha] \otimes_{\iota(K)} K_\lambda$. Hence β has order 5, $\beta + \beta^{-1} = \iota(\vartheta)$ with $\vartheta^2 + \vartheta - 1 = 0$, and the characteristic polynomial of β on $V_\lambda(J_0)$ is its minimal polynomial $x^2 - \vartheta x + 1$ over $\iota(K)$. As β multiplies $U\,dU/V$ by ζ^2, $\iota(\vartheta)$ multiplies $U\,dU/V$, and so du/v (Proposition 2.2(b)), by $\zeta^2 + \zeta^{-2}$.

(c) In the same way, with $\varrho \in \overline{\mathcal{K}}$, $\varrho^3 = -1/t$, the isomorphisms $h_Y(U,V) = (-1/(YU), -V/(Y^3U^3))$ from $C_{1,Y}$ to C_{-1/Y^3}, the curve $V^2 = U^6 + 10U^3 + 12\varrho U + 5$ over $\overline{K}[[\varrho]]$ (discriminant $2^{12}3^6 5^5(1+\varrho^3)^2$) with special fiber $C_{1,0}$, and the maps $a_\mu(U,V) = (\mu U, V)$, $\mu \in \mu_3$, from $C_{1,Y}$ to $C_{1,\mu^2 Y}$, with $h_{\mu^2 Y} \circ a_\mu = h_Y$: $g \in G_\mathcal{K}$ acts on $T_\lambda(J_\infty)$ through $(U,V) \mapsto (\xi U, V)$, $\xi = g(\varrho)/\varrho$, which for $g = \gamma_\infty$ is α or α^2 (as $\overline{K}((1/t))(\varrho)/\overline{K}((1/t))$ is totally ramified of degree 3) for the automorphism α of Lemma 2.5(b). By its proof, $\mathbb{Q}[\iota(\varphi), \alpha] \cong K(\sqrt{-3})$, so $\alpha \notin \iota(K)$, and $V_\lambda(J_\infty)$ is free of rank one over $K(\sqrt{-3}) \otimes_K K_\lambda$; so the characteristic polynomial is $x^2 + x + 1$.

(b) As in the proof of Lemma 5.17 at $t = 1$ (where $\lambda = \sqrt{5}$; the argument does not depend on λ), C_t has a regular projective model over $\overline{K}[[t-1]]$ whose special fiber is a rational curve with two ordinary double points of thickness 1, and γ_1 acts on $T_\ell(J_t)$ by $1 + N$, where $N\colon T_\ell/T_\ell^I \to T_\ell^I$ is given by the unimodular monodromy pairing $\pm\,\mathrm{diag}(1,1)$, so is an isomorphism [4, Exp. IX, 9.2, 12.3–12.4 and Théorème 12.5]. As N commutes with ι, it induces an isomorphism $T_\lambda/T_\lambda^I \to T_\lambda^I$ of $\mathcal{O}_\lambda$-modules, both free of rank one (T_λ^I is saturated). If e_1 generates T_λ^I and e_2 generates T_λ modulo T_λ^I, then $\rho(\gamma_1)e_2 = e_2 + ne_1$ with $n \in \mathcal{O}_\lambda^\times$; take the basis (ne_1, e_2).

Finally, $\det \rho(\gamma_i) = 1$ for each i, and γ_0, γ_1 generate Π topologically. □

Lemma S1.6. *Let $p \equiv \pm 1 \pmod 5$, $p \geq 11$, $\mathfrak{p} \mid p$, and H as above.*

(a) $\mathcal{X}_H(\mathfrak{p})$ is a smooth projective geometrically connected curve over K, and t has degree $[G : H]$: $p+1$ for B, $p(p+1)/2$ for N_{sp} and $p(p-1)/2$ for N_{ns}.

(b) For a field $L \supseteq K$ and $t_0 \in \mathcal{U}(L)$, $\mathcal{X}_H(\mathfrak{p})$ has an L-point over t_0 if and only if $\bar{\rho}_{t_0}(G_L)$ lies in a conjugate of H.

(c) A K-point of $\mathcal{X}_{\mathrm{sp}}^+$ over $t_0 \in \mathcal{U}(K)$ is the same as a K-rational divisor $R + R'$ with $R \neq R'$ in the fiber of $\mathcal{X}_B$ over t_0, that is, a K-point of $\mathrm{Sym}^2 \mathcal{X}_B$ of this form.

(d) For $t_0 \in \mathcal{U}(\mathbb{Q})$, whether $\mathcal{X}_H(\mathfrak{p})$ has a K-point over t_0 does not depend on the choice of $\mathfrak{p} \mid p$.

Proof. Monodromy. Let $\gamma_0, \gamma_1, \gamma_\infty$ be as in Lemma S1.5, and A, B_0, C their actions on the fiber $\mathbb{F}_p^2$ of $\mathcal{J}[\mathfrak{p}]$ at a geometric generic point, the reductions modulo $\mathfrak{p}$ of $\rho(\gamma_0)$, $\rho(\gamma_1)$, $\rho(\gamma_\infty)$ for $\lambda = \mathfrak{p}$. By Lemma S1.5, $\det A = \det B_0 = 1$, $\operatorname{tr} A = \vartheta \in \mathbb{F}_p$ with $\vartheta^2 + \vartheta - 1 = 0$, $B_0 \neq 1$ is unipotent, and $AB_0 = C^{-1}$ has characteristic polynomial $x^2 + x + 1$, so $\operatorname{tr}(AB_0) = -1$. As $\vartheta(\vartheta+1) = 1$, $\vartheta \neq -1$, so the fixed line of B_0 is not fixed by A, else $\operatorname{tr}(AB_0) = \operatorname{tr} A$. In a basis where this line is ∞, the powers of B_0 give all upper unipotent matrices ($\mathbb{F}_p$ is a prime field), and conjugating them by A followed by a translation, the composite sending ∞ to 0, gives the lower ones; so $\langle A, B_0 \rangle = \mathrm{SL}_2(\mathbb{F}_p)$, and this is the image of $\pi_1(\mathcal{U}_{\overline{K}})$. The images of A, B_0, C in $\mathrm{PGL}_2(\mathbb{F}_p)$ have orders $5, p, 3$: as $(x^2 - \vartheta x + 1)(x^2 + (1+\vartheta)x + 1) = x^4 + x^3 + x^2 + x + 1$ and $p \neq 3, 5$, the eigenvalues of A, resp. C, are distinct primitive fifth, resp. cube, roots of unity.

(a) The geometric components of $\mathcal{Y}_H(\mathfrak{p})$ are the orbits on G/H of a group containing $\mathrm{SL}_2(\mathbb{F}_p)$, which is transitive exactly when $\det H = \mathbb{F}_p^\times$; this holds for B, N_{sp} and N_{ns}. (b) The L-points over t_0 are the cosets xH with $\bar{\rho}_{t_0}(G_L) \subseteq xHx^{-1}$. (c) The cosets of N_{sp} are the unordered pairs of distinct lines. (d) For $\gamma \in G_{\mathbb{Q}} \smallsetminus G_K$, $\gamma \circ [\alpha] = [\sigma(\alpha)] \circ \gamma$ (Proposition 2.2(c)), so for $t_0 \in \mathcal{U}(\mathbb{Q})$, γ maps $J_{t_0}[\mathfrak{p}]$ onto $J_{t_0}[\mathfrak{p}']$ and conjugates the image of G_K for $\mathfrak{p}$ to that for $\mathfrak{p}'$. □

Lemma S1.7. *Let $p \geq 7$ split in K, $\mathfrak{p} \mid p$, and M/K a quadratic extension unramified at $\mathfrak{p}$. Let W be a two-dimensional $\mathbb{F}_p$-representation of G_K that is odd, finite flat at $\mathfrak{p}$, with determinant $\bar{\chi}_p$, irreducible, and with $W \cong W \otimes \theta_M$. Then $W \otimes \overline{\mathbb{F}}_p$ has exactly two G_M-stable lines, exchanged by $G_K \smallsetminus G_M$, and:*

(a) if $\mathfrak{p}$ splits in M, both are defined over $\mathbb{F}_p$, so the image of G_M lies in C_{sp} and that of G_K in N_{sp} but not in C_{sp};

(b) if $\mathfrak{p}$ is inert in M, they are conjugate lines over $\mathbb{F}_{p^2}$, so the image of G_M lies in C_{ns} and that of G_K in N_{ns}; moreover the inertia group at $\mathfrak{p}$ fixes no line of W and acts transitively on the $p+1$ lines.

Proof. As W is absolutely irreducible, an isomorphism $W \to W \otimes \theta_M$ over $\overline{\mathbb{F}}_p$ has scalar square and is not scalar; its eigenlines are preserved by G_M, exchanged by $G_K \smallsetminus G_M$, and are the only G_M-stable lines (else W would be reducible), so they are rational or conjugate over $\mathbb{F}_{p^2}$.

The inertia group $I_{\mathfrak{p}}$ lies in G_M. By Raynaud's theorem [11, Théorème 3.4.3 and Corollaire 3.4.4] (see [12, §2.8]) and $\det = \bar{\chi}_p$, $W|^{\mathrm{ss}}_{I_{\mathfrak{p}}} \otimes \overline{\mathbb{F}}_p$ is $1 \oplus \bar{\chi}_p$ or $\omega_2 \oplus \omega_2^p$, ω_2 a fundamental character of level 2. (b) If $\mathfrak{p}$ is inert in M, then on a decomposition group at $\mathfrak{p}$, W is induced from $\psi_1|_{G_{\mathbb{Q}_{p^2}}}$, where ψ_1 is the character of G_M on one of the two G_M-stable

lines, so on $I_\mathfrak{p}$ it is $\omega_2^k \oplus \omega_2^{kp}$; this is not $\{1, \bar\chi_p\}$, so it is $\{\omega_2, \omega_2^p\}$, not $\mathbb{F}_p$-valued. Hence no $\mathbb{F}_p$-line is $I_\mathfrak{p}$-stable, the lines are not rational, and the image of $I_\mathfrak{p}$ in $\mathrm{PGL}_2(\mathbb{F}_p)$, cyclic of order $p+1$ and acting freely on $\mathbb{P}^1(\mathbb{F}_p)$, is transitive. (a) If $\mathfrak{p}$ splits in M, W is a sum of two characters of $G_{\mathbb{Q}_p}$, which are $\mathbb{F}_p$-valued on $I_\mathfrak{p}$, so $W|^{\mathrm{ss}}_{I_\mathfrak{p}} = 1 \oplus \bar\chi_p$; an element of C_{ns} with eigenvalues $1, \bar\chi_p(\gamma) \in \mathbb{F}_p$ is scalar, so the lines are rational. □

Proposition S1.8. *Assume the hypotheses of Theorem 6.21, and let (X, Y, Z) be a primitive solution, with $t = -X^5/Y^3 \in \mathcal{U}(\mathbb{Q})$.*

(a) In class (E), $\mathcal{X}_B$ has a K-point over t, whose line has character θ_{15} or $\theta_{15}\bar\chi_p$.

(b) In class (Z) with $p \equiv 31, 41 \pmod{60}$ and in class (T) with $p \equiv 19, 31 \pmod{60}$, $\mathcal{X}^+_{\mathrm{sp}}$ has a K-point over t; equivalently, there are two distinct points R, R^c of $\mathcal{X}_B$ over t, defined over the field of complex multiplication M of the class and exchanged by $\mathrm{Gal}(M/K)$, so that $R + R^c$ is a K-point of $\mathrm{Sym}^2 \mathcal{X}_B$.

(c) In class (Z) with $p \equiv 19, 29 \pmod{60}$ and in class (T) with $p \equiv 29, 41 \pmod{60}$, $\mathcal{X}^+_{\mathrm{ns}}$ has a K-point over t. In these cases the fiber of $\mathcal{X}_B$ over t is a single closed point of degree $p+1$, so $\mathcal{X}_B$ and $\mathrm{Sym}^2 \mathcal{X}_B$ have no K-point over t.

Proof. (a) is Theorem 6.21 with Lemma S1.6(b). In (Z), (T), Lemma S1.7 applies to $W = J_t[\mathfrak{p}]$ (finite flat at $\mathfrak{p}$ by Theorem 2.8, irreducible by Proposition 6.4, and $W \cong W \otimes \theta_M$). $\mathfrak{p}$ splits in $\mathbb{Q}(\zeta_5)$, resp. $K(\sqrt{-3})$, if and only if $p \equiv 1 \pmod 5$, resp. $\pmod 3$, and $(p \bmod 5, p \bmod 3)$ is $(4,1), (4,2), (1,1), (1,2)$ for $p \equiv 19, 29, 31, 41 \pmod{60}$. (b) The two G_M-stable lines give points of $\mathcal{X}_B$ over M, exchanged by $\mathrm{Gal}(M/K)$; apply Lemma S1.6(c). (c) The image of G_K lies in N_{ns}, and G_K is transitive on the $p+1$ lines by Lemma S1.7(b). □

Proposition S1.9. *Let $p \equiv \pm 1 \pmod 5$, $p \geq 11$, and put $\delta_5 = \pm 1$ according as $p \equiv \pm 1 \pmod 5$ and $\delta_3 = \pm 1$ according as $p \equiv \pm 1 \pmod 3$. Then*

$$g(\mathcal{X}_B) = \frac{7p - 15 - 12\delta_5 - 10\delta_3}{30},$$
$$g(\mathcal{X}^+_{\mathrm{sp}}) = 1 + \frac{(p+1)(7p-15)}{60} - \frac{1+\delta_5}{5} - \frac{1+\delta_3}{6},$$
$$g(\mathcal{X}^+_{\mathrm{ns}}) = 1 + \frac{(p-1)(7p-15)}{60} - \frac{1-\delta_5}{5} - \frac{1-\delta_3}{6},$$

and $g(\mathcal{X}^+_{\mathrm{sp}}) = g(\mathcal{X}_B) + g(\mathcal{X}^+_{\mathrm{ns}})$. For $p = 139, 149, 151, 281$ (the smallest $p \geq 139$ in each class modulo 60) these genera are $(32, 2236, 2204)$, $(35, 2571, 2536)$, $(34, 2640, 2606)$, $(65, 9175, 9110)$.

Proof. Riemann–Hurwitz gives $2g - 2 = -2d + \sum_{P \in \{0,1,\infty\}} (d - \#\text{orbits at } P)$, $d = [G : H]$, and an element of prime order ℓ with F fixed points on G/H has $(d + (\ell-1)F)/\ell$ orbits. The local monodromies have orders $5, p, 3$ (proof of Lemma S1.6); those of order 5, 3 are split semisimple if $p \equiv 1$ modulo 5, resp. 3, and non-split otherwise, and that of order p is unipotent. On lines a split, non-split, unipotent element fixes 2, 0, 1; an element of odd prime order fixes a pair of lines only if it fixes both, so on pairs of distinct (resp. Frobenius-conjugate) lines a split (resp. non-split) semisimple element fixes 1 and the others none. Substituting gives the formulas. □

J_t has *potentially toric reduction* at v if over a finite extension of K_v its Néron special fiber contains a nontrivial torus; as the toric rank is 0 or 2 ($\mathcal{O}_K$ acts on the character group), this means not potentially good ("potentially multiplicative" in [10, Proposition 7.4]).

Lemma S1.10. *Assume the first sentence of* (T2.1) *and* (P7.4)(a) *at* $\ell = 3$*. For every primitive solution with* $p \geq 7$*,* J_t *has potentially toric reduction at some prime of* K*.*

Proof. If $Z = \pm 1$, then $|X|, |Y| \geq 2$ (if $|X| = 1$ then $Y^3 \in \{0, \pm 2\}$; if $|Y| = 1$ then $X = 0$), so $|X|^5$ and $|Y|^3$ are consecutive integers greater than 1, against Mihăilescu's theorem [8]. Let $\ell \mid Z$. If $\ell \neq 3, 5$, the reduction above ℓ is multiplicative (Theorem 2.8); if $\ell = 3$, then $v_3(t-1) > 3$ and (P7.4)(a) applies; if $\ell = 5$, Lemma 2.19(b) applies. □

Definition S1.11. Let $p \equiv 19, 29, 31, 41 \pmod{60}$ and $\mathfrak{p} \mid p$. Put $N_Z = N_{\mathrm{sp}}$ if $p \equiv 1 \pmod 5$ and $N_Z = N_{\mathrm{ns}}$ if $p \equiv 4 \pmod 5$; put $N_T = N_{\mathrm{sp}}$ if $p \equiv 1 \pmod 3$ and $N_T = N_{\mathrm{ns}}$ if $p \equiv 2 \pmod 3$. Hypothesis (M_p) is the statement: *for each* $H \in \{B, N_Z, N_T\}$*, no point* $R \in \mathcal{X}_H(\mathfrak{p})(K)$ *has* $t(R) \in \mathbb{Q} \smallsetminus \{0, 1\}$ *with* $J_{t(R)}$ *of potentially toric reduction at some prime of* K*.* By Lemma S1.6(d), (M_p) does not depend on the choice of $\mathfrak{p}$.

The curves are listed in Table S1. By Lemma S1.6(c), for $p \equiv 31 \pmod{60}$ (M_p) concerns only $\mathcal{X}_B$ and $\mathrm{Sym}^2 \mathcal{X}_B$ (points $R + R'$ with $R \neq R'$, $t(R) = t(R')$). For the other three residues of p modulo 60 it involves the non-split Cartan curve $\mathcal{X}_{\mathrm{ns}}^+$, for which the corresponding question is open even for elliptic curves over $\mathbb{Q}$.

TABLE S1. The curves in (M_p).

p mod 60	class (E)	class (Z)	class (T)
19	$\mathcal{X}_B$	$\mathcal{X}_{\mathrm{ns}}^+$	$\mathcal{X}_{\mathrm{sp}}^+$
29	$\mathcal{X}_B$	$\mathcal{X}_{\mathrm{ns}}^+$	$\mathcal{X}_{\mathrm{ns}}^+$
31	$\mathcal{X}_B$	$\mathcal{X}_{\mathrm{sp}}^+$	$\mathcal{X}_{\mathrm{sp}}^+$
41	$\mathcal{X}_B$	$\mathcal{X}_{\mathrm{sp}}^+$	$\mathcal{X}_{\mathrm{ns}}^+$

Theorem S1.12. *Assume* (LL) *and* (EL)*, together with* (T2.1), (P7.4)(a) *at* $\ell = 3$ *and* (Ex_0)*. Let* $p \equiv 19, 29, 31, 41 \pmod{60}$ *with* $p \geq 139$*. If* (M_p) *holds, then* $X^5 + Y^3 = Z^p$ *has no primitive solution; equivalently,* $x^5 + y^p + z^3 = 0$ *has no nontrivial primitive solution.*

Proof. By Theorem 6.21 and Proposition S1.8, a primitive solution gives $R \in \mathcal{X}_H(\mathfrak{p})(K)$ with $t(R) = t \in \mathbb{Q} \smallsetminus \{0, 1\}$, where $H = B$, N_Z, N_T in (E), (Z), (T); by Lemma S1.10 J_t has potentially toric reduction somewhere, against (M_p). □

Remark S1.13. The condition on potentially toric reduction removes only fibers with potentially good reduction everywhere, such as fibers with complex multiplication. We have not settled how (M_p) compares with [10, Conjecture 1].

S1.3. **The primes above p in the classes with complex multiplication.**

Proposition S1.14. *Assume* (LL) *and* (EL)*, together with* (T2.1), (P7.4)(a) *at* $\ell = 3$ *and* (Ex_0)*. Let* $p \equiv 19, 29, 31, 41 \pmod{60}$ *with* $p \geq 139$*, and let* (X, Y, Z) *be a primitive solution in class (Z) or (T) of Theorem 6.21, with* M *the field of complex multiplication attached to that class in Theorem 6.21, such that* $\mathfrak{p}$ *is inert in* M *(that is, class (Z) with* $p \equiv 19, 29 \pmod{60}$*, or class (T) with* $p \equiv 29, 41 \pmod{60}$*). Then* $p \nmid Z$*,* J_t *has good reduction at* p*, and the Frobenius polynomial of its reduction over* $\mathbb{F}_p$ *is* $(T^2 + p)^2$*.*

Proof. The proof of Lemma S1.7(b) applies to $W = J_t[\mathfrak{p}]$ at $\mathfrak{p}$ and at $\mathfrak{p}'$ (W is finite flat at both, and $\mathfrak{p}'$ is inert in M as $M/\mathbb{Q}$ is Galois): inertia at $\mathfrak{p}$ and at $\mathfrak{p}'$ fixes no line of W.

If $p \mid Z$, the reduction at $\mathfrak{p}$ is multiplicative of toric rank 2 (Theorem 2.8), and the $\mathfrak{p}$-torsion of the torus [2, Chapter III] would be such a line. So $p \nmid Z$, $\operatorname{disc} \mathcal{F}_{X,Y} = 2^{12}3^6 5^5 Z^{2p}$ is a unit at p, and the reduction at $\mathfrak{p}$, $\mathfrak{p}'$ is good. Let $a_{\mathfrak{l}} \in \mathcal{O}_K$ ($\mathfrak{l} \in \{\mathfrak{p}, \mathfrak{p}'\}$) be the trace of $\mathrm{Frob}_{\mathfrak{l}}$ on $V_\ell(J_t)$ over $K \otimes \mathbb{Q}_\ell$. By [6] the crystalline Frobenius of the $\mathfrak{p}$-part (of height 2, dimension 1) of the p-divisible group of the reduction at $\mathfrak{l}$ has characteristic polynomial $T^2 - \iota_{\mathfrak{p}}(a_{\mathfrak{l}})T + p$, where $\iota_{\mathfrak{p}} \colon \mathcal{O}_K \to \mathcal{O}_{K_{\mathfrak{p}}} = \mathbb{Z}_p$ is the completion at $\mathfrak{p}$. If $\iota_{\mathfrak{p}}(a_{\mathfrak{l}})$ were a unit, the multiplicative part of that p-divisible group would give such a line; so $a_{\mathfrak{p}}, a_{\mathfrak{p}'} \in \mathfrak{p}$. As $a_{\mathfrak{p}'} = \sigma(a_{\mathfrak{p}})$ (Proposition 2.2(c)), $a_{\mathfrak{p}} \in p\mathcal{O}_K$; its real conjugates have absolute value at most $2\sqrt{p} < p/2$, so $|N(a_{\mathfrak{p}})| < p^2$ and $a_{\mathfrak{p}} = 0$. The Frobenius polynomial over $\mathbb{F}_p$ is $N_{K/\mathbb{Q}}(T^2 + p) = (T^2 + p)^2$. □

Let S_p be the set of $\tau \in \mathbb{F}_p \smallsetminus \{0, 1\}$ such that $\operatorname{Jac} C_\tau$ (with $C_\tau \colon v^2 = 5u^6 - 12u^5 + 10\tau u^3 + \tau^2$, smooth over $\mathbb{F}_p$) has Frobenius polynomial $(T^2+p)^2$ over $\mathbb{F}_p$, and for $c, a \in \mathbb{F}_p^\times$ let $D_0(c) \colon v^2 = u^6 + 12cu$ and $D_\infty(a) \colon v^2 = u^6 + 10au^3 + 5a^2$.

Proposition S1.15. *Let $p \equiv 19, 29, 41 \pmod{60}$ with $139 \le p \le 5000$; there are 117 such primes, 39 in each class.*

(a) S_p is empty for 98 of them: 32 with $p \equiv 19$, 32 with $p \equiv 29$ and 34 with $p \equiv 41 \pmod{60}$. For the other 19, S_p is given in Table S2; it has at most three elements.

(b) $\operatorname{Jac} D_0(c)$ has Frobenius polynomial $(T^2 + p)^2$ for every $c \in \mathbb{F}_p^\times$ if $p \equiv 4 \pmod 5$ and for no c if $p \equiv 1 \pmod 5$; $\operatorname{Jac} D_\infty(a)$ has it for every $a \in \mathbb{F}_p^\times$ if $p \equiv 2 \pmod 3$ and for no a if $p \equiv 1 \pmod 3$.

Consequently, under the assumptions of Proposition S1.14, for each of these p with $S_p = \emptyset$, a primitive solution in a class of Proposition S1.14 has $t \equiv 0$ or $\infty \pmod p$, the parameter of a trivial solution: $p \mid X$ in class (Z) for $p \equiv 19$, $p \mid Y$ in class (T) for $p \equiv 41$, and $p \mid XY$ in classes (Z), (T) for $p \equiv 29 \pmod{60}$.

TABLE S2. The nonempty sets S_p for $p \equiv 19, 29, 41 \pmod{60}$, $139 \le p \le 5000$.

p	S_p	p	S_p	p	S_p
281	{132}	2549	{1058}	4159	{599}
389	{292}	2659	{155, 604, 1738}	4219	{1223}
821	{606}	2969	{2836}	4241	{3883}
1229	{744}	3041	{1997}	4481	{494}
1459	{985}	3319	{2699, 3187}	4639	{1634}
1889	{1799}	3449	{597}		
1949	{1065}	4099	{2249}		

Proof. Parts (a) and (b) are a machine computation (item C21 of Table S3): for every τ the trace of Frobenius is computed exactly by a character sum, and every τ with trace 0 is tested with `hyperellcharpoly`; for (b) one c modulo fifth powers and one a modulo cubes suffice, as $D_0(c) \cong D_0(cb^5)$ and $D_\infty(a) \cong D_\infty(ab^3)$ for $b \in \mathbb{F}_p^\times$.

Consequence: by Proposition S1.14 the reduction of $C_{X,Y}$ at p has Frobenius polynomial $(T^2 + p)^2$; it is $C_{\bar{t}}$ with $\bar{t} \in \mathbb{F}_p \smallsetminus \{0, 1\}$ if $p \nmid XY$ (Lemma 2.1), $D_0(\bar{Y})$ if $p \mid X$, and $D_\infty(\bar{X})$ if $p \mid Y$. Apply (a) and (b). □

TABLE S3. Computations used in the Supplement.

Item	Establishes	Used in	Software
C20	The genera of $\mathcal{X}_B$, $\mathcal{X}_{\mathrm{sp}}^+$, $\mathcal{X}_{\mathrm{ns}}^+$ from the permutation actions of the monodromy elements of orders 5, p, 3 and the Riemann–Hurwitz formula, for $p = 11, 19, 29, 31, 41, 139, 149, 151, 199, 211, 269, 271, 281$; they agree with the formulas	Proposition S1.9	Python, SymPy, SageMath
C21	For the 117 primes $p \equiv 19, 29, 41 \pmod{60}$, $139 \le p \le 5000$: the sets S_p, the number of τ with trace of Frobenius 0, and the verdicts of Proposition S1.15(b)	Proposition S1.15, Table S2	Python, PARI/GP, C
C22	For the newform Θ of the remark on [10, Remark 7.11] in §2.5 of the paper, $M = K(\sqrt{-3})$, $L = \mathbb{Q}(\zeta_{15})$ and each of the 174 primes λ of L above the auxiliary primes $7 \le \ell \le 389$ of [10]: $\psi(N_{L/M}\lambda) = -\chi_\lambda(5)^5 J(\chi_\lambda^2, \chi_\lambda^8)$, with J and $\chi_\lambda(5)$ by enumerating $\mathcal{O}_L/\lambda$, and $\psi(N_{L/M}\lambda) = r^{f(\lambda/\mathfrak{l})}$, $\mathfrak{l}$ the prime of K below λ, and, with $a_{\mathfrak{l}}(\Theta)$ computed from a Brandt module and agreeing with [13], r a root of $x^2 - a_{\mathfrak{l}}(\Theta)x + N\mathfrak{l}$ chosen so that $(r^{f(\lambda/\mathfrak{l})}) = \prod_{a\in\{1,2,4,8\}} \sigma_a(\lambda)$, $\sigma_a(\zeta_{15}) = \zeta_{15}^a$ (the two roots are equal when $\ell \equiv 7, 13 \pmod{15}$ and give the same value when $\mathfrak{l}$ is inert in M); $\prod_{a\in\{1,2,4,8\}} \sigma_a(\lambda)$ is the ideal of J at the 64 primes of degree 1; $(\mathcal{O}_L/\mathfrak{p}_3^4\mathfrak{p}_5)^\times \cong \mathbb{Z}/720 \times \mathbb{Z}/72 \times \mathbb{Z}/9 \times (\mathbb{Z}/3)^7$ is generated modulo sixth powers by generators of these primes (already those above $\ell \le 100$) and four units	§2.5 of the paper	PARI/GP, Python

Remark S1.16. Proposition S1.15 never excludes $t \equiv 0$ or $\infty \pmod{p}$, and says nothing for $p > 5000$, $p \equiv 31 \pmod{60}$, or the classes in which $\mathfrak{p}$ splits in M.

S2. COMPUTATIONS

The computations used in the Supplement, and item C22, used in §2.5 of the paper, are listed in Table S3, numbered after those of the paper (Appendix A), whose conventions they follow. They were made with PARI/GP 2.17.3 [14], Python 3.12 (with NumPy, SymPy, cypari2 and python-flint), SageMath 10.9 [15] and, for item C21, the C programming language. The programs, their outputs and the commands that run them are in [5].

Repetitions. A repetition is a computation by a separately written program from the same mathematical statements and the same models of the curves; a second PARI/GP program shares PARI's library functions with the first, and repetitions marked Sage use no part of Sage that calls PARI.

C20 A second Python program (other generators, another model of $\mathbb{F}_{p^2}$, a count of fixed points), with the formulas derived in SymPy; and Sage, with the permutation actions computed in GAP, for all the primes.

C21 Three methods: Python with character sums and `hyperellcharpoly`; Python with Hasse–Witt matrices; and C, without PARI, by point counts over $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$. For $p \le 1500$, `hyperellcharpoly` at every τ.

C22 Python with python-flint, without PARI: the primes from the factorizations of Φ_{15} modulo ℓ, J from a table of discrete logarithms, generators of the primes found by lattice reduction and certified by their norms, and an explicit model of $(\mathcal{O}_L/\mathfrak{p}_3^4\mathfrak{p}_5)^\times$

modulo sixth powers; and a second PARI/GP program, with `bnrinit`, by which the classes of the primes above $\ell \leq 17$ already generate the ray class group modulo 2 and modulo 3.

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Toronto, Canada
Email address: manvirjaswal@icloud.com